

Mauzam Shafi Bhat

Durham, North Carolina, USA | mb948@duke.edu | +1 (919) 282-8855 | [linkedin.com/in/mauzamshafibhat](https://www.linkedin.com/in/mauzamshafibhat) | github.com/MAUZAMSHAFIBHAT

EDUCATION

Duke University

USA

Master of Science in Computer Science

Aug 2025 – May 2027 (Expected)

Relevant Coursework: Software Development Studio, Distributed Systems, Database Systems, Operating Systems, Algorithms, Computer Networks, Applied ML & Data Science

SRM Institute of Science and Technology

India

Bachelor of Technology in Electronics and Computer Engineering (CGPA: 8.53/10)

Sep 2020 – Jun 2024

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Machine Learning

SKILLS

Languages: Python, C, C++, SQL, JavaScript, HTML/CSS

Backend & Web: Django, Django REST Framework, FastAPI, Flask, Celery, React, REST APIs, JWT, Microservices

Cloud & Data: AWS (S3, Lambda, SageMaker), Docker, Docker Compose, Linux, Git, PostgreSQL, MySQL, Redis

ML/AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, YOLO, Mask R-CNN, NetworkX, OpenAI API

Testing/QA: Pytest, Unit and Integration Testing, API Contract Testing, Regression Testing, Code Coverage Analysis

EXPERIENCE

Software Development Intern

Oct 2024 – Dec 2024

Infosys

- Built Django REST APIs powering real-time dashboards; profiled latency bottlenecks and cut API response time by 30% (200 ms to 140 ms) using Redis caching and database query optimization.
- Fine-tuned ML waste categorization models and integrated them into the Django backend, raising accuracy from 68% to 85%.
- Moved long-running data processing jobs to asynchronous Celery task queues, increasing pipeline throughput by 35%.

Machine Learning Intern

Mar 2023 – Jul 2023

Eduskills

- Built a churn prediction pipeline on AWS SageMaker; parallelized S3 data loading to cut batch processing time by 40%.
- Designed AWS Lambda ETL workflows processing 500K+ records daily and automated model retraining on fresh data; the churn model achieved an F1-score of 0.87.

Data Science Intern

Apr 2022 – Jun 2022

Tech Analogy

- Developed time-series forecasting models with XGBoost on 100K+ historical records, achieving an R^2 score of 0.92.
- Automated recurring SQL reporting pipelines, reducing the team's manual analysis time by 60%.

PROJECTS

AI-Powered Code Analysis Platform

Python, FastAPI, OpenAI API (GPT-5), Docker, Microservices

- Engineered a multi-agent code review backend in FastAPI with 5 specialized GPT-5 agents (Style, Security, Performance, Testing, Clean Code), including a Testing agent that achieved 60–80% defect detection accuracy across 40+ projects.
- Built an asynchronous REST API and an automated test suite with 70%+ code coverage; fixing the defects it surfaced cut failed API requests by 40%.

Music Streaming Platform (Spotify Clone)

React, Vite, Django REST Framework, PostgreSQL, JWT, Docker Compose

- Built a full-stack music streaming app (React 18 + Vite frontend, Django REST Framework backend) with 15+ REST endpoints for auth, songs, playlists, and social features; secured with JWT and containerized with Docker Compose.
- Wrote 60+ automated tests covering API contracts, authentication flows, and edge cases, reaching 96.7% code coverage.

Network Topology Reconstruction System

Python, YOLO, Mask R-CNN, NetworkX

- Built a computer vision pipeline that reconstructs network topologies from cloud architecture diagrams (AWS, Azure, GCP) using YOLO object detection and Mask R-CNN segmentation; achieved 85% detection precision and 80% reconstruction accuracy.
- Modeled detections as a NetworkX graph and applied graph algorithms to infer links, reducing topology mapping time by 25%.

ACTIVITIES & LEADERSHIP

Milan Cultural Fest, SRM University

2023

Convener

- Coordinated 150+ student organizers across logistics, event ticketing, and attendee check-in.